# EXPERIMENT NO. 14

Configure and Verify NTP Operating in Client and Server Mode

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## Configure and Verify NTP Operating in Client and Server Mode

**AIM:**To configure and verify **Network Time Protocol (NTP)** operating in client and server mode using **Cisco Packet Tracer**.

NTP is an application-layer protocol used to synchronize the clocks of devices on a TCP/IP network.

### OBJECTIVES

- To create a network topology for NTP configuration.
- To configure an NTP server.
- To configure Router0 and Router1 as NTP clients.
- To verify connectivity between the routers and NTP server.
- To verify that the clocks of Router0 and Router1 are synchronized with the NTP server.
- To check NTP synchronization using show clock detail and show ntp status.

### THEORY

**Network Time Protocol (NTP)** is an application-layer protocol used to synchronize the time and date of computers and network devices over a network. It follows a **client-server model**.

An **NTP server** provides the reference time to NTP clients. In this experiment, the server has IP address **209.165.200.225**, while Router0 and Router1 are configured as NTP clients.

The NTP server distributes its time information to the routers so that their clocks remain synchronized. NTP is useful because synchronized time is required for many network applications, logging, troubleshooting and security-related operations.

| S.NO | Device | Interface | IPv4 Address | Subnet Mask |
|---|---|---|---|---|
| 1. | Server-PT | nil | 209.165.200.225 | 255.255.255.0 |
| 2. | router0 | GigabitEthernet0/0 | 209.165.200.226 | 255.255.255.0 |
| 3. | router1 | GigabitEthernet0/0 | 209.165.200.227 | 255.255.255.0 |

### SOFTWARE / MATERIALS REQUIRED

- Cisco Packet Tracer 8.x
- Computer System

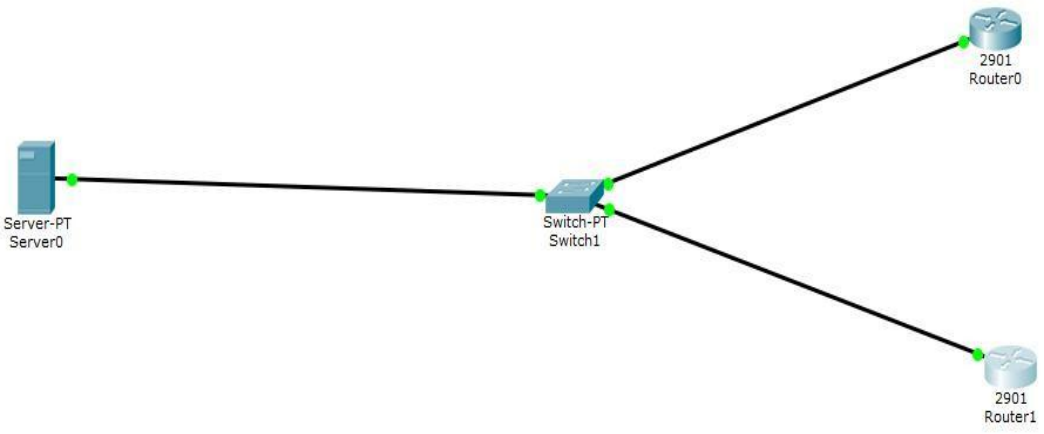
## IMPORTANT PROGRAM STEPS

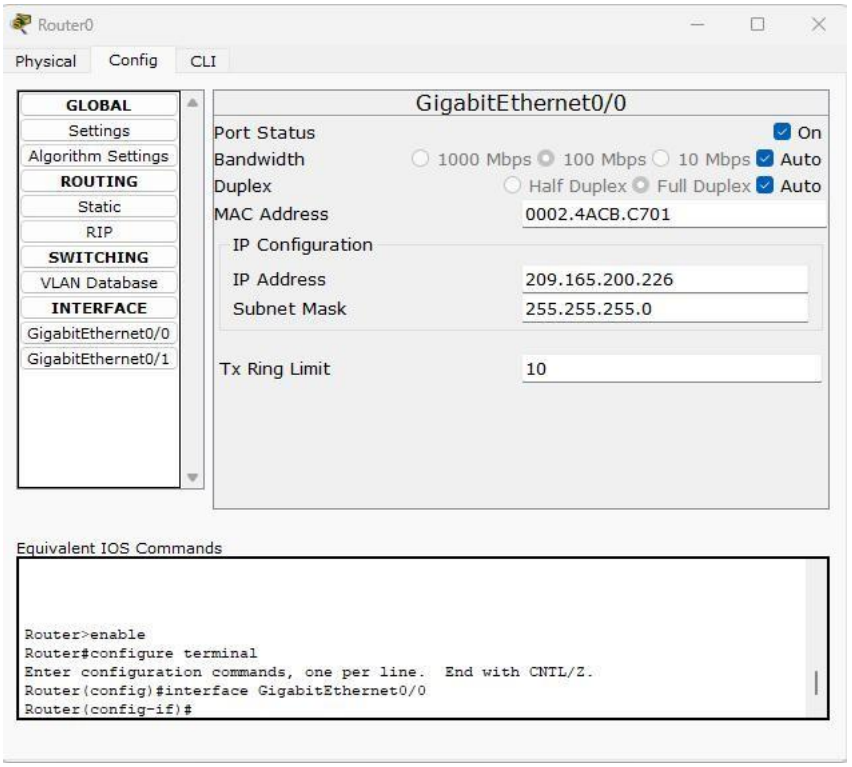
- **Server0** acts as the NTP server.
- **Router0** acts as an NTP client.
- **Router1** acts as an NTP client.
- The switch provides connectivity between the devices.
- Both routers obtain their time from the NTP server.

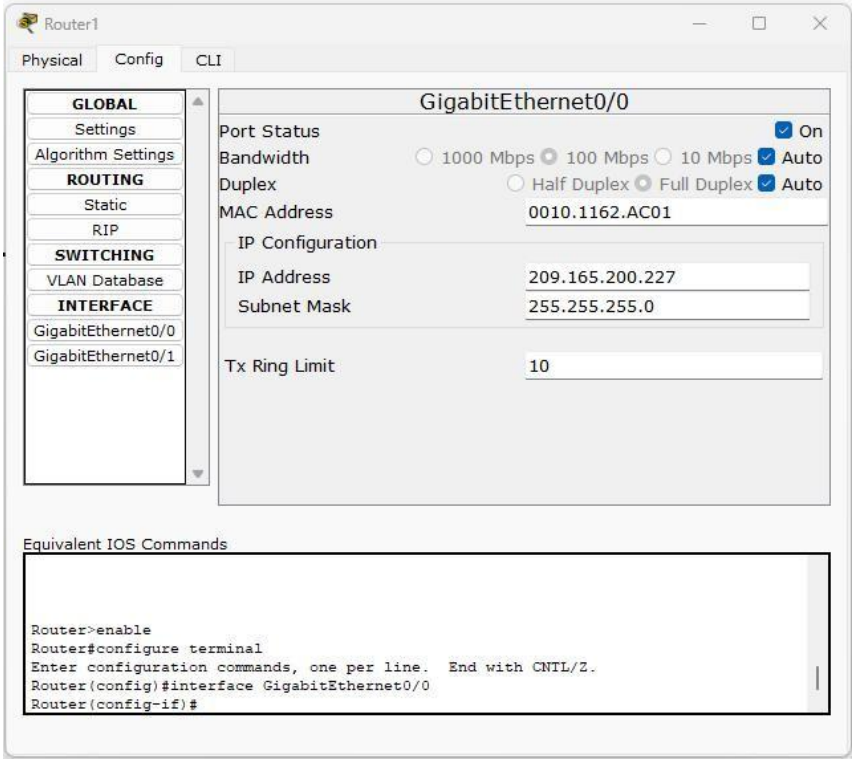
## PROCEDURE

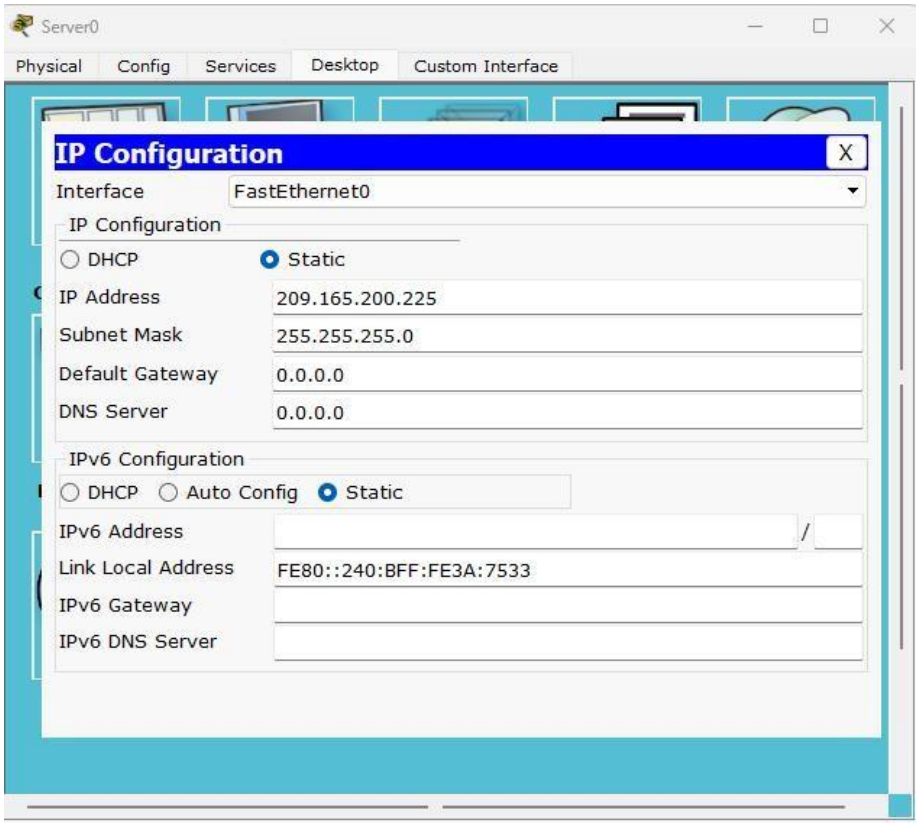
- Configure IP addresses on Server0, Router0, and Router1.
- Connect Server0, Router0, and Router1 through the switch.
- Verify connectivity between the NTP server and routers using `ping`.
- Configure Router0 as an NTP client using the NTP server address.
- Configure Router1 as an NTP client using the NTP server address.
- Verify NTP synchronization using `show clock detail`.
- Verify NTP status using `show ntp status`.
- Confirm that Router0 and Router1 are synchronized with the NTP server.

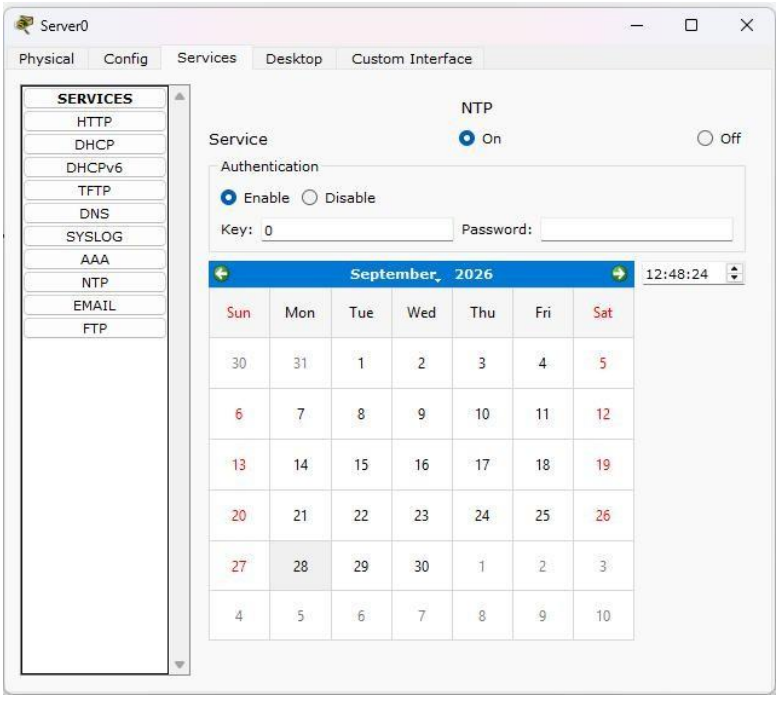
## RESULT

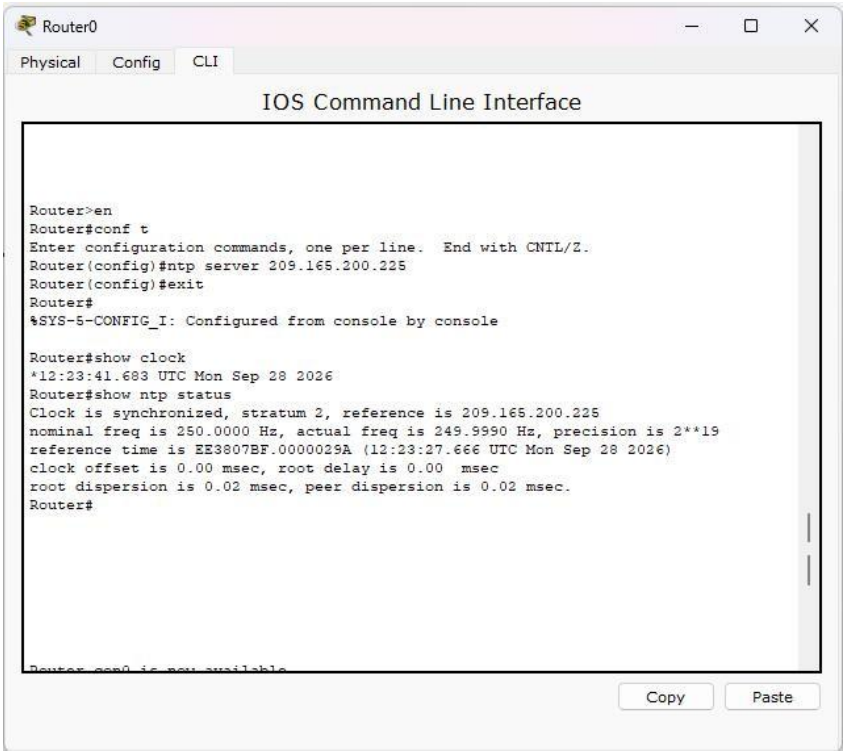


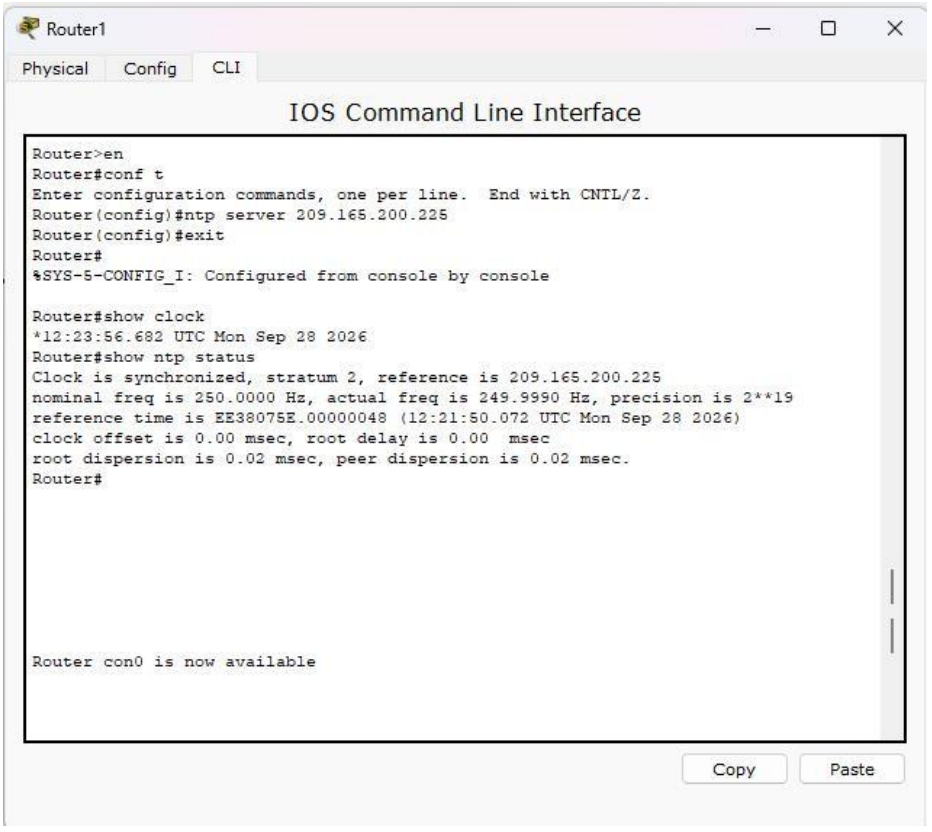

Router0
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0002.4ACB.C701
IP Configuration
IP Address
209.165.200.226
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#


Router1
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0010.1162.AC01
IP Configuration
IP Address
209.165.200.227
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#


Server0
Physical
Config
Services
Desktop
Custom Interface
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IP Address
209.165.200.225
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::240:BFF:FE3A:7533
IPv6 Gateway
IPv6 DNS Server


Server0
Physical
Config
Services
Desktop
Custom Interface
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
NTP
Service
On
Off
Authentication
Enable
Disable
Key:
0
Password:
September 2026
12:48:24
Sun
Mon
Tue
Wed
Thu
Fri
Sat


Router0
Physical
Config
CLI
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ntp server 209.165.200.225
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show clock
*12:23:41.683 UTC Mon Sep 28 2026
Router#show ntp status
Clock is synchronized, stratum 2, reference is 209.165.200.225
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**19
reference time is EE3807BF.0000029A (12:23:27.666 UTC Mon Sep 28 2026)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 0.02 msec, peer dispersion is 0.02 msec.
Router#
Copy
Paste


Router1
Physical
Config
CLI
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ntp server 209.165.200.225
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show clock
*12:23:56.682 UTC Mon Sep 28 2026
Router#show ntp status
Clock is synchronized, stratum 2, reference is 209.165.200.225
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**19
reference time is EE38075E.00000048 (12:21:50.072 UTC Mon Sep 28 2026)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 0.02 msec, peer dispersion is 0.02 msec.
Router#
Router con0 is now available
Copy
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## CONCLUSION

The NTP client-server configuration was successfully implemented in Cisco Packet Tracer. The NTP server was configured and both Router0 and Router1 were configured as NTP clients. Connectivity was verified using the ping command, and synchronization was verified using show clock detail and show ntp status. Both routers successfully synchronized their clocks with the NTP server 209.165.200.225.